

THEORY



Training Program 01

Advanced NLP & Large Language Models (LLMs)



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Training Program 01

M01: Text Preprocessing

M02: Word and Sentence Embeddings

M03: Transformer Architectures

M04: Entity-Centric NLP

M05: Applied NLP Tasks

M06: Multimodal NLP

M07: LLM Fundamentals

M08: Prompt Engineering

M09: Fine-Tuning & PEFT

M10: RAG & Real-World Projects



MODULE M01

TEXT PREPROCESSING

CHAPTER M03

Transformer Architectures



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3.1 From RNNs to the Transformer Revolution

3.2 The Transformer: Attention Is All You Need

3.3 Encoder vs Decoder Architectures

3.4 Encoder-Only Models (BERT Family)

3.5 Decoder-Only Models (GPT Family)

3.6 Fine-Tuning Strategies and Best Practices

3.7 Summary and The Future of LLMs



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Chapter 3.1 From RNNs to the Transformer Revolution

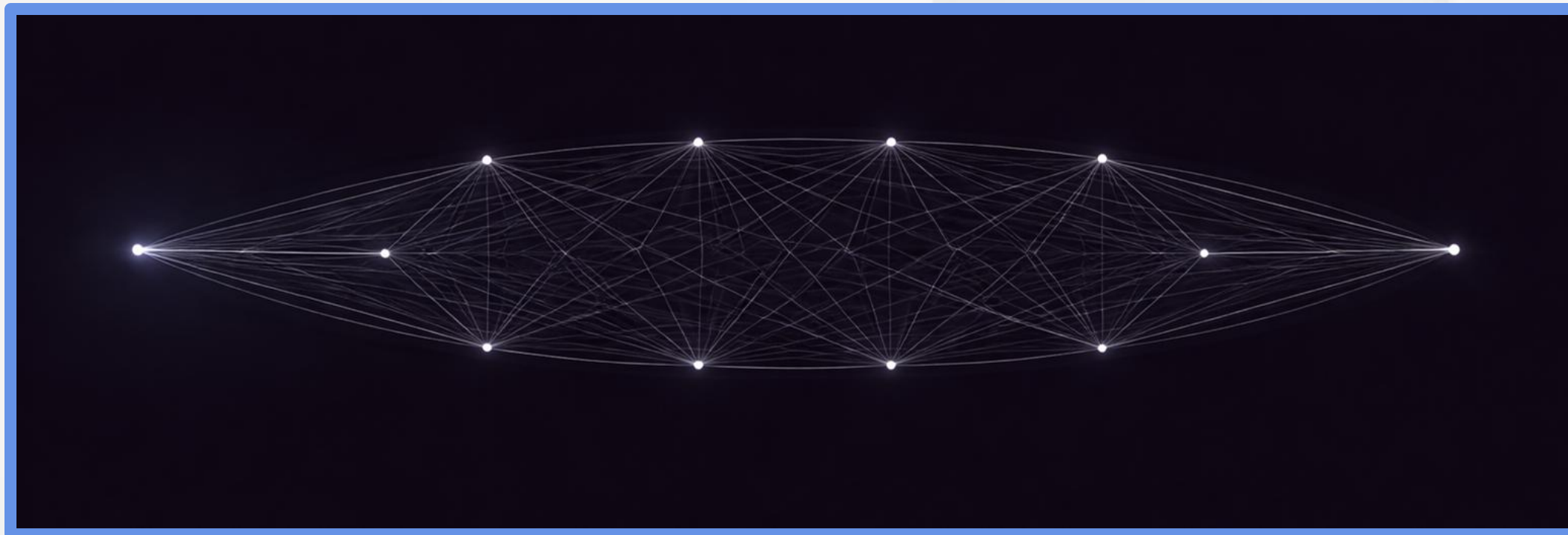
- RNNs process sequences token by token.
- LSTMs and GRUs improved memory handling.
- Dominant approach for sequence modeling.
- Fundamentally sequential computation.



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Chapter 3.1 From RNNs to the Transformer Revolution

- Tokens attend to each other directly.
- No sequential bottleneck.
- Full parallelization on GPUs.
- Long-range relationships become trivial.



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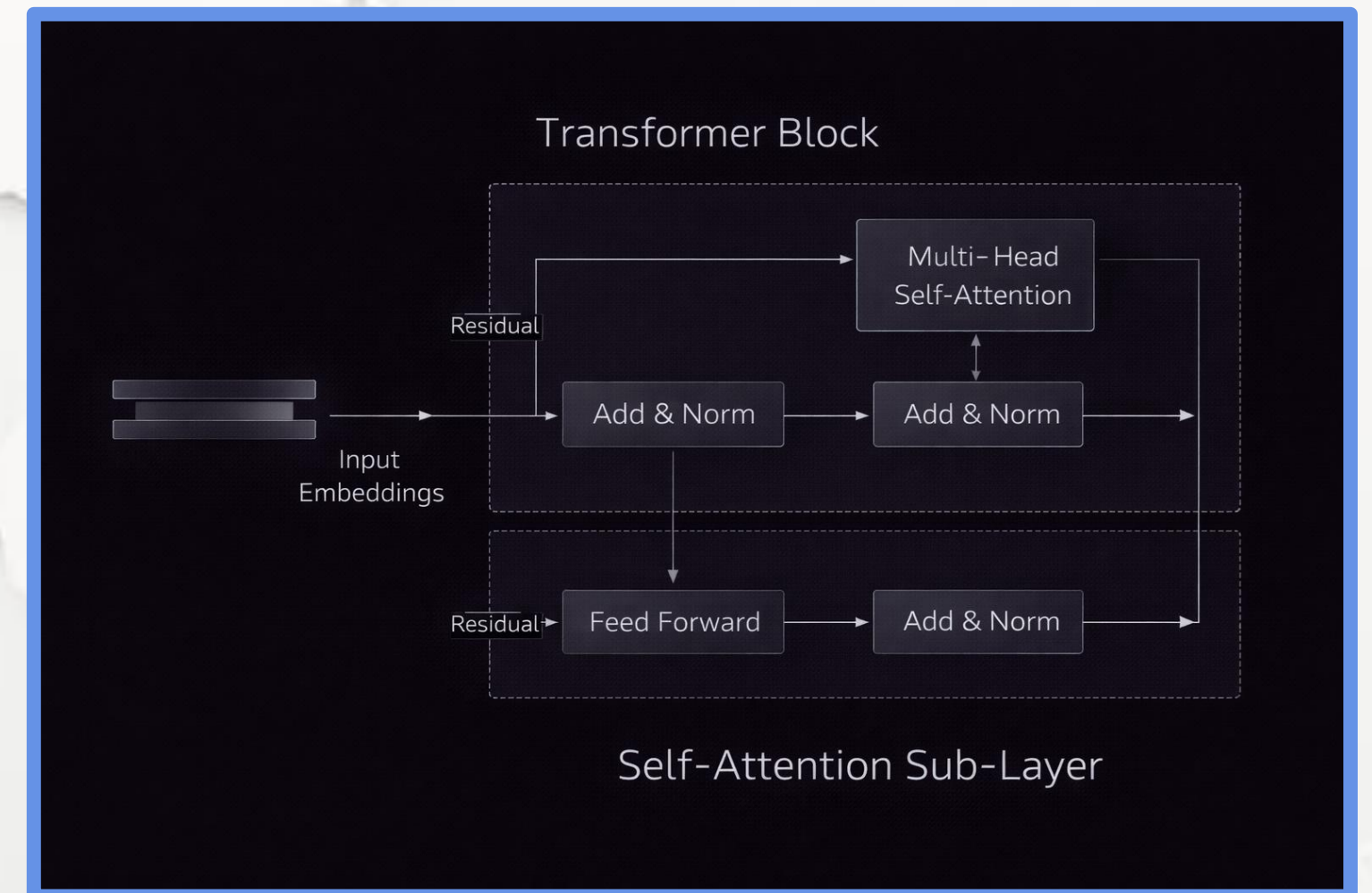


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Chapter 3.2 The Transformer: Attention Is All You Need

Self-Attention: Query, Key and Value

- Each token generates Query, Key and Value vectors.
- Queries match against all Keys.
- Values are weighted by attention scores.
- Output is a context-aware representation.

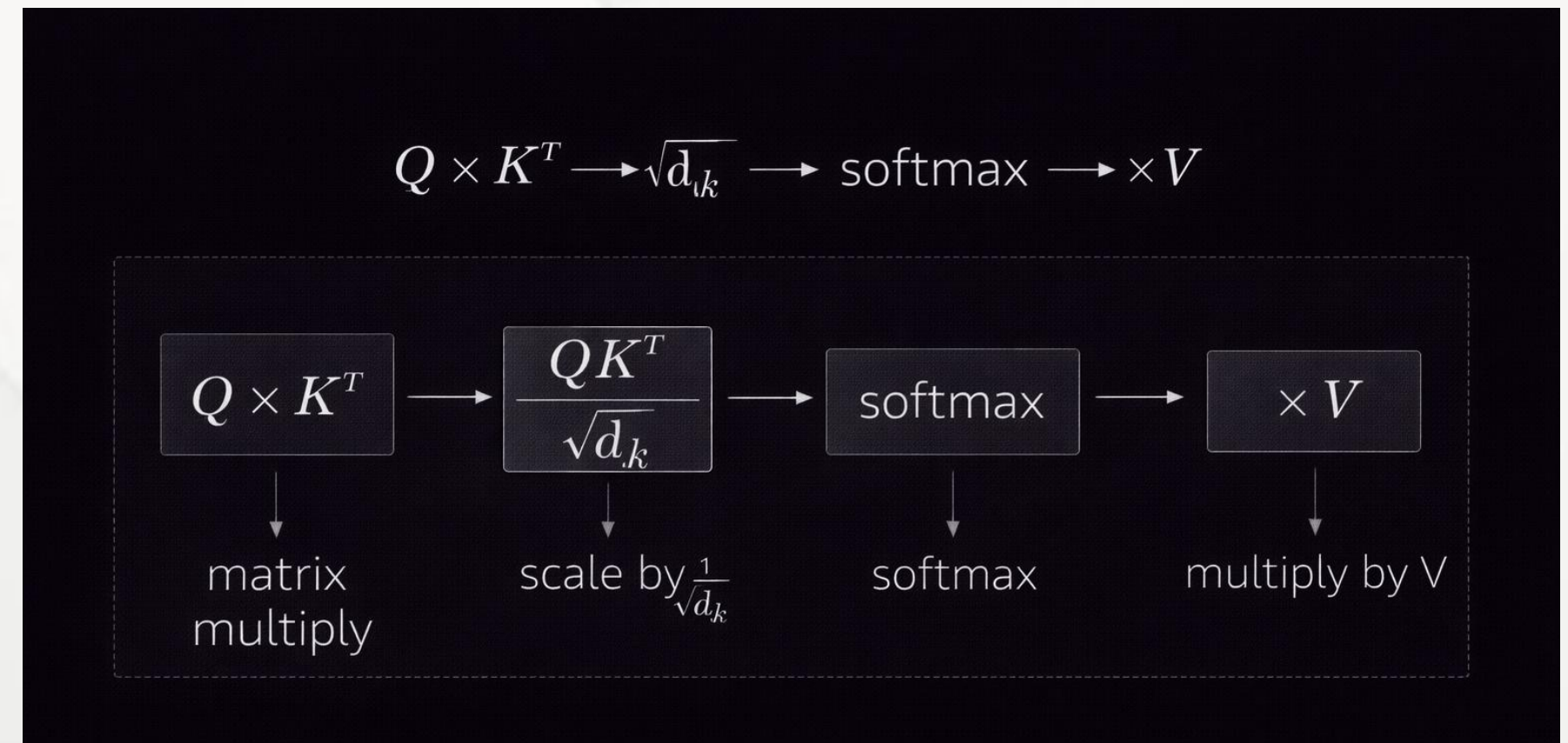


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Chapter 3.2 The Transformer: Attention Is All You Need

Self-Attention: Query, Key and Value

- Dot product measures token compatibility.
- Scores scaled by $\sqrt{d_k}$ for stability.
- Softmax converts scores to probabilities.
- Weighted sum produces attention output.

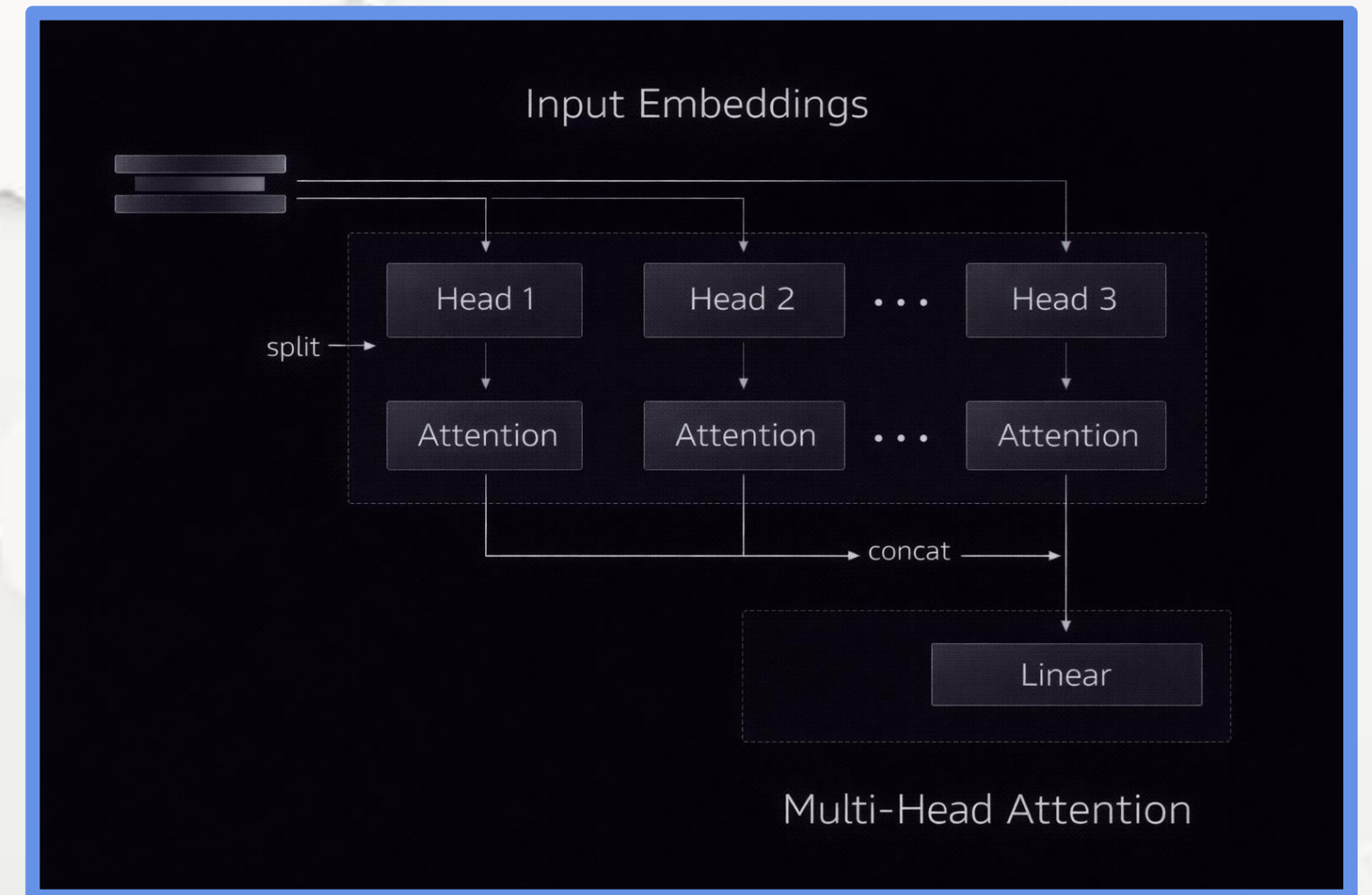


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Chapter 3.2 The Transformer: Attention Is All You Need

Multi-Head Attention

- Dot product measures token compatibility.
- Scores scaled by $\sqrt{d_k}$ for stability.
- Softmax converts scores to probabilities.
- Weighted sum produces attention output.



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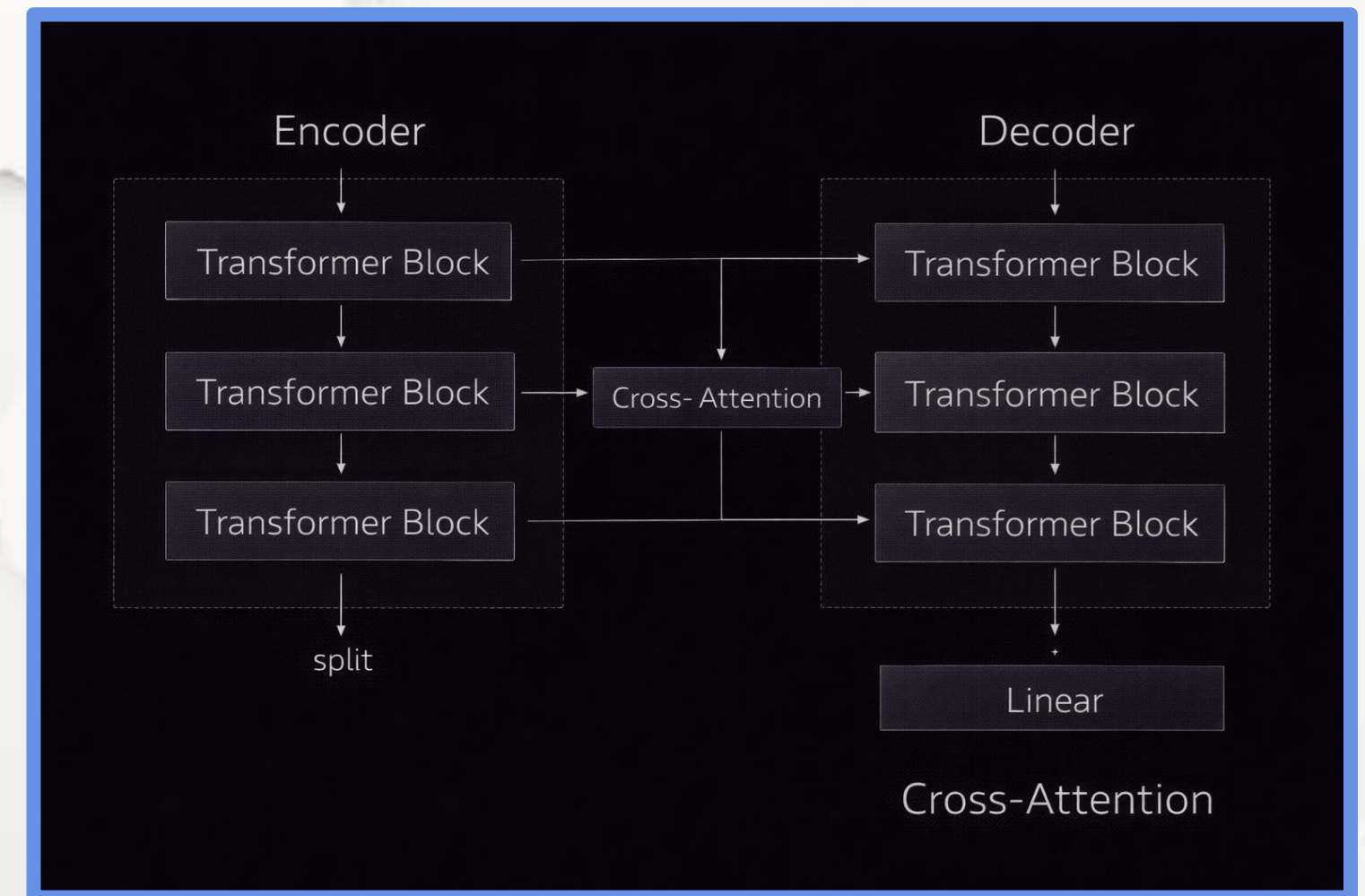


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Chapter 3.3 Encoder vs Decoder Architectures

Sequence-to-Sequence Transformer Overview

- Original Transformer uses Encoder–Decoder structure.
- Encoder processes input sequence.
- Decoder generates output sequence.
- Cross-attention connects both stacks.



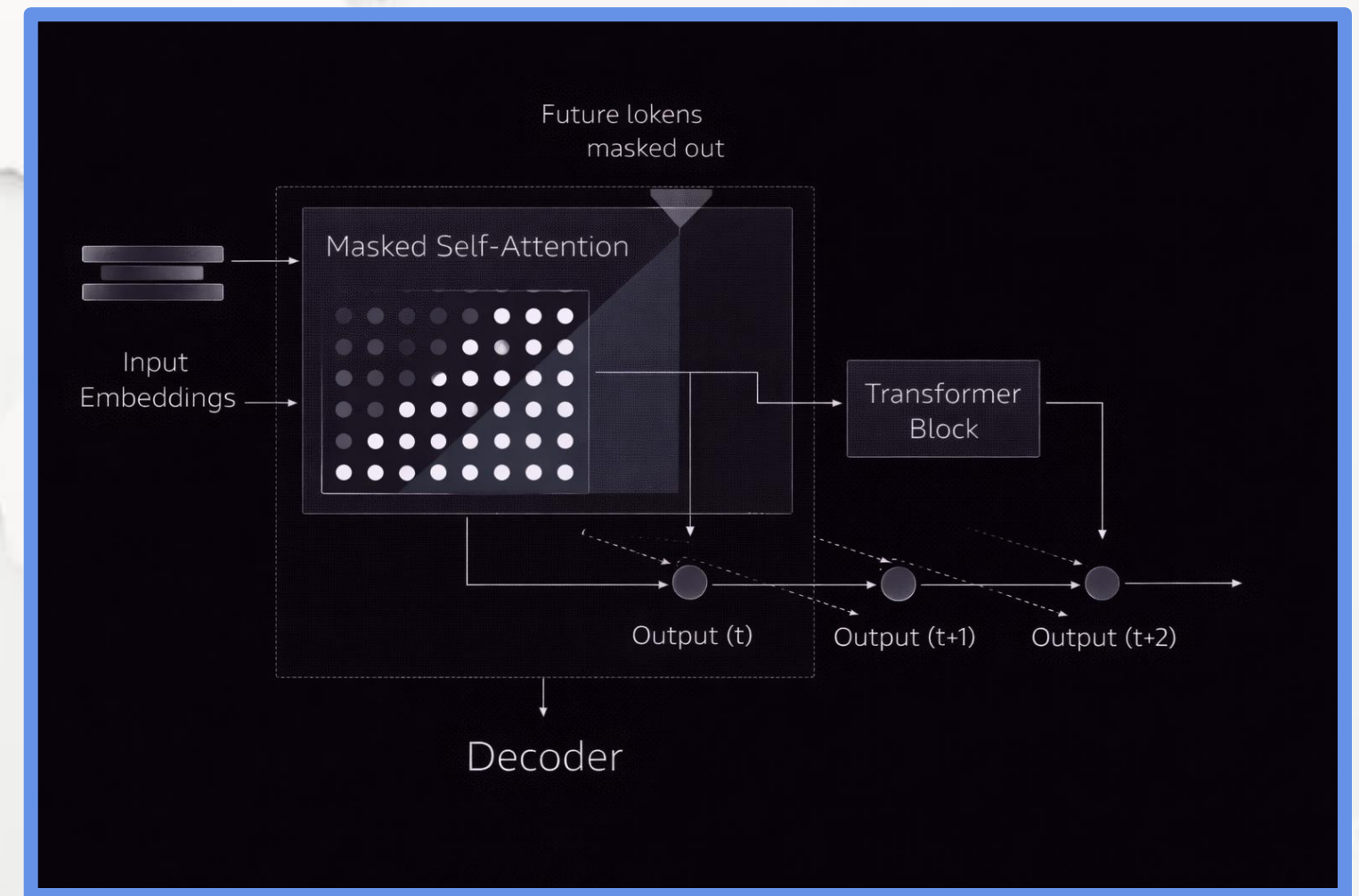
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Chapter 3.3 Encoder vs Decoder Architectures

Encoder-Only Architecture (Understanding)

- Full bidirectional self-attention.
- Tokens see left and right context.
- Produces rich contextual representations.
- Optimized for understanding tasks.



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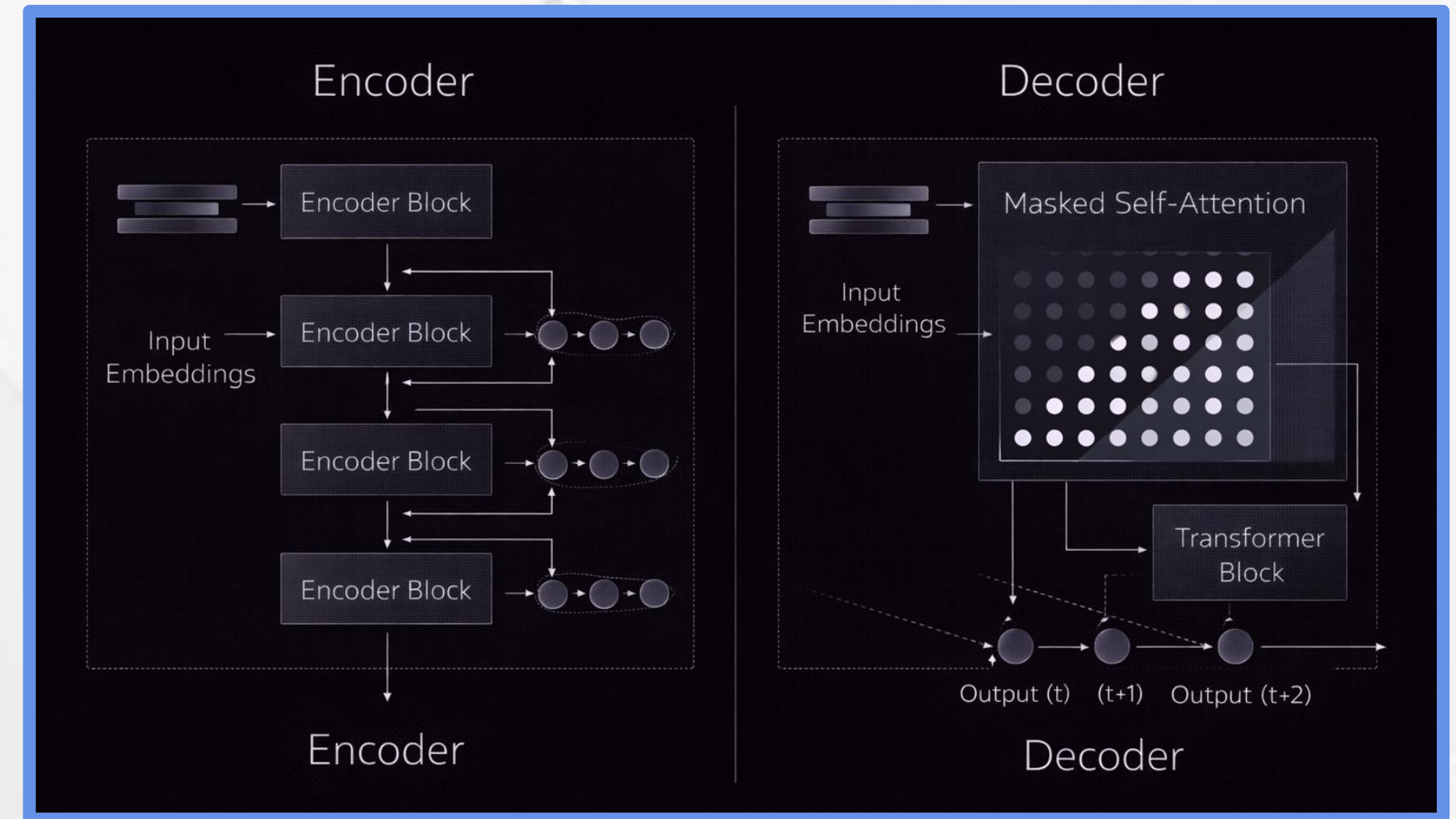
Chapter 3.3 Encoder vs Decoder Architectures

Decoder-Only Architecture (Generation)

- Causal (masked) self-attention.
- Tokens only attend to past positions.
- Auto-regressive generation.
- Optimized for text generation.

Architectural Comparison: Encoder vs Decoder

- Causal (masked) self-attention.
- Tokens only attend to past positions.
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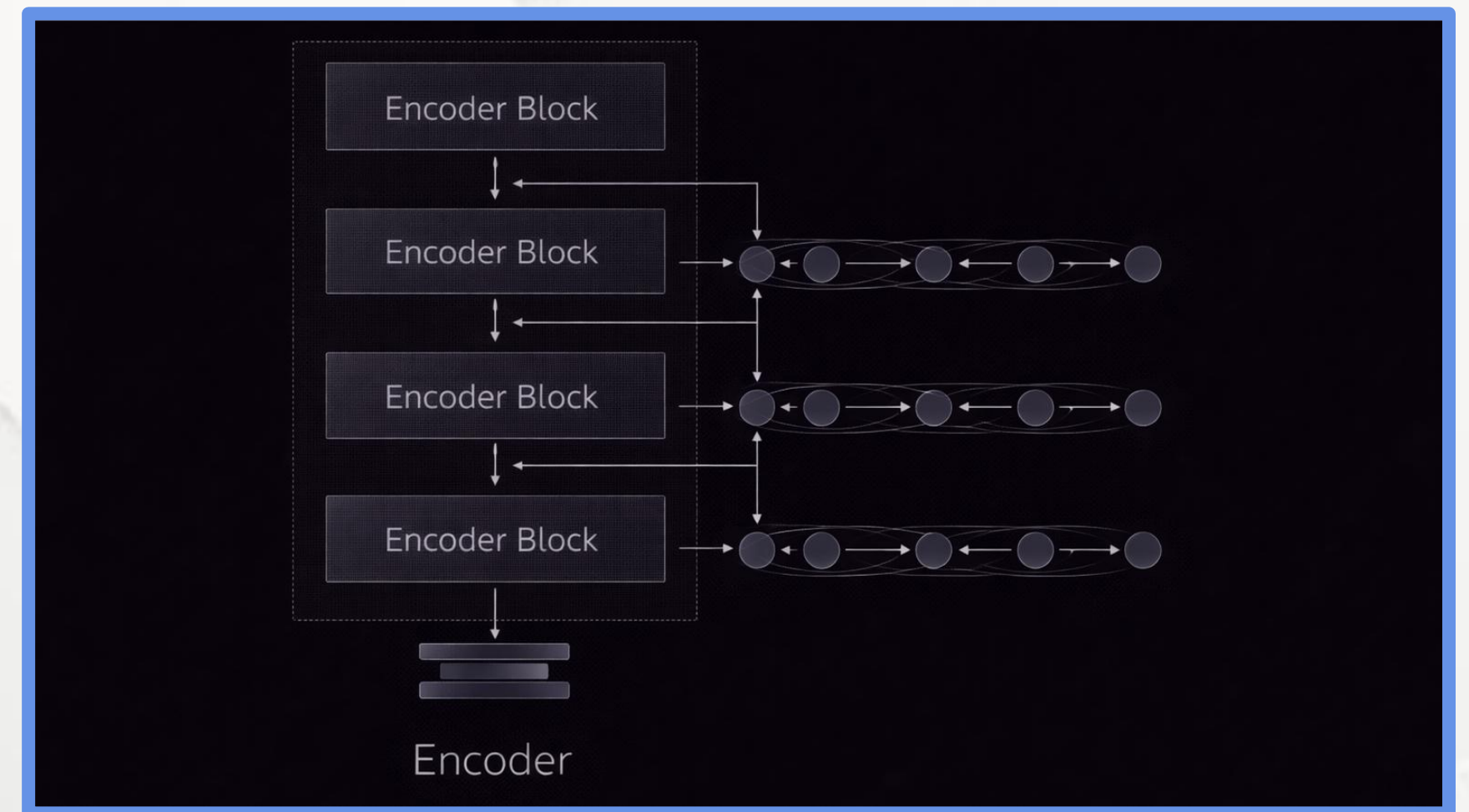


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Chapter 3.4 Encoder-Only Models

What Is an Encoder-Only Transformer?

- Uses only the encoder stack.
- Full bidirectional self-attention.
- Builds deep contextual representations.
- Designed for understanding tasks.



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Chapter 3.4 Encoder-Only Models

BERT: Bidirectional Context Modeling

- Tokens attend left and right simultaneously.
- Meaning depends on full sentence context.
- Same word gets different vectors per context.
- Solves polysemy effectively.



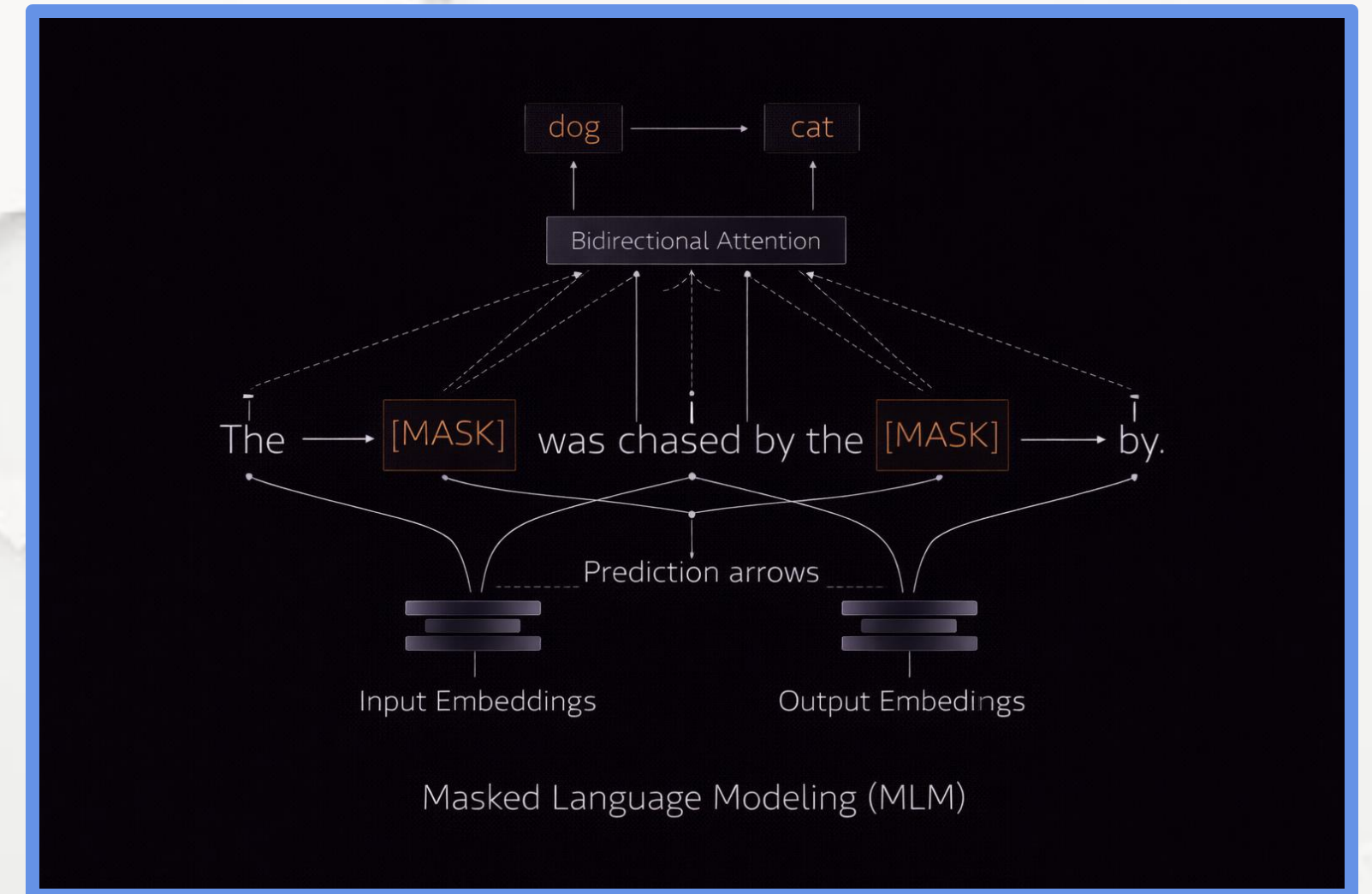
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Chapter 3.4 Encoder-Only Models

Typical Use Cases of Encoder-Only Models

- Text classification.
- Sentiment analysis.
- Named Entity Recognition.
- Semantic similarity scoring.



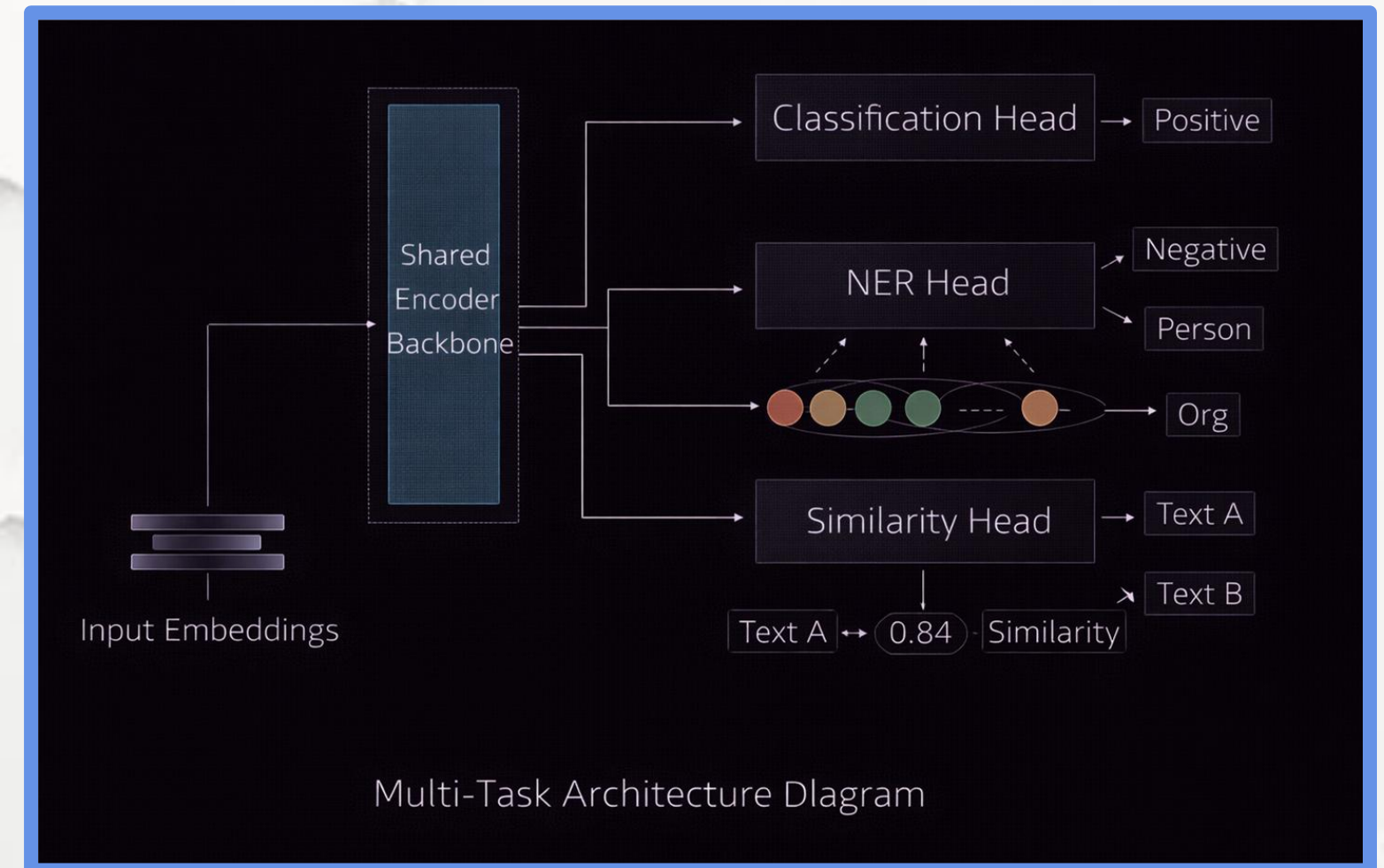
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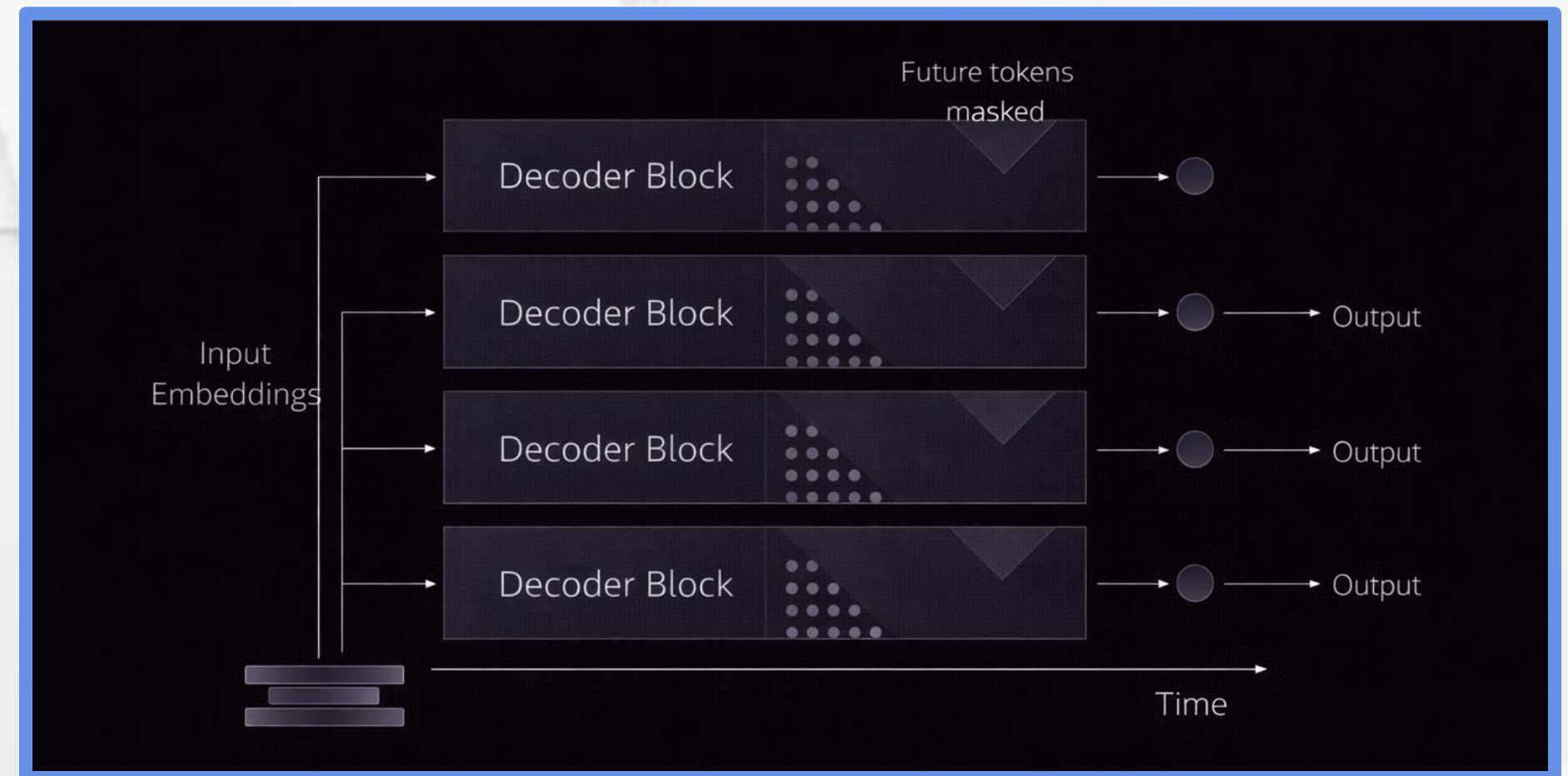


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Chapter 3.5 Decoder-Only Models

What Is a Decoder-Only Transformer?

- Uses only the decoder stack.
- No bidirectional attention.
- Generates text token by token.
- Optimized for sequence generation.



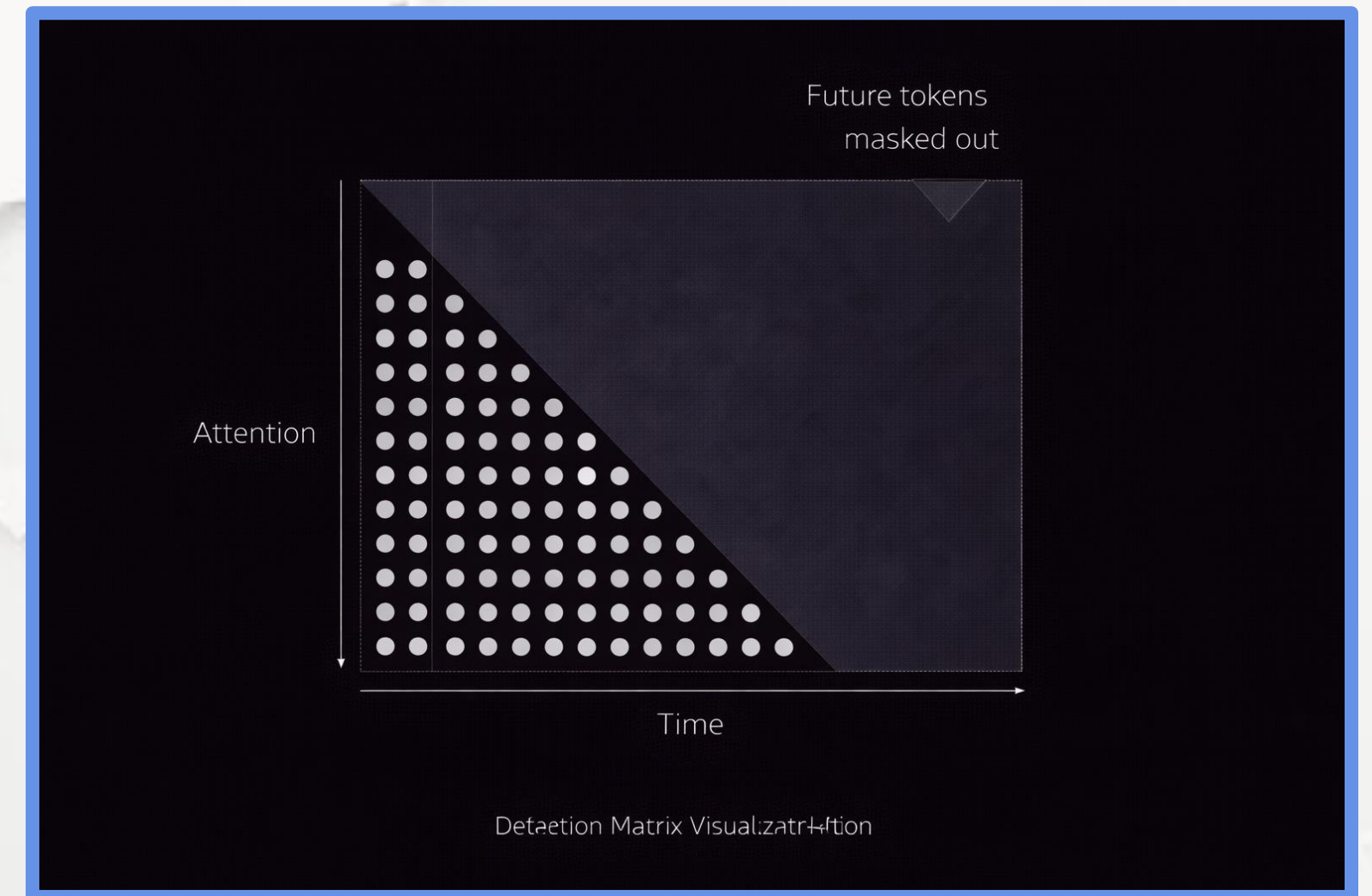
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Chapter 3.5 Decoder-Only Models

Causal (Masked) Self-Attention

- Tokens can only attend to past tokens.
- Future positions are masked.
- Prevents information leakage.
- Enables auto-regressive learning.



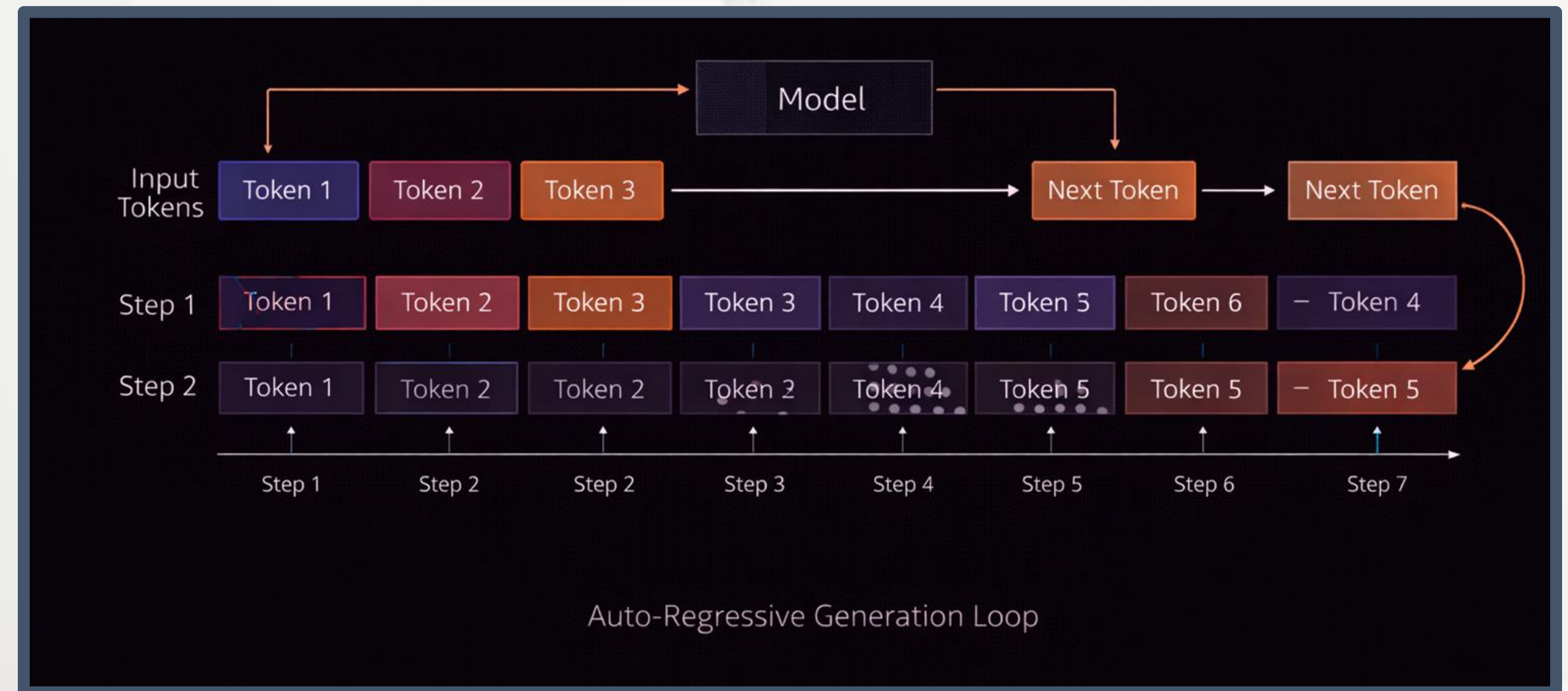
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Chapter 3.5 Decoder-Only Models

Auto-Regressive Generation Loop

- Predict next token probability.
- Append generated token to input.
- Repeat until stop condition.
- Text emerges sequentially.



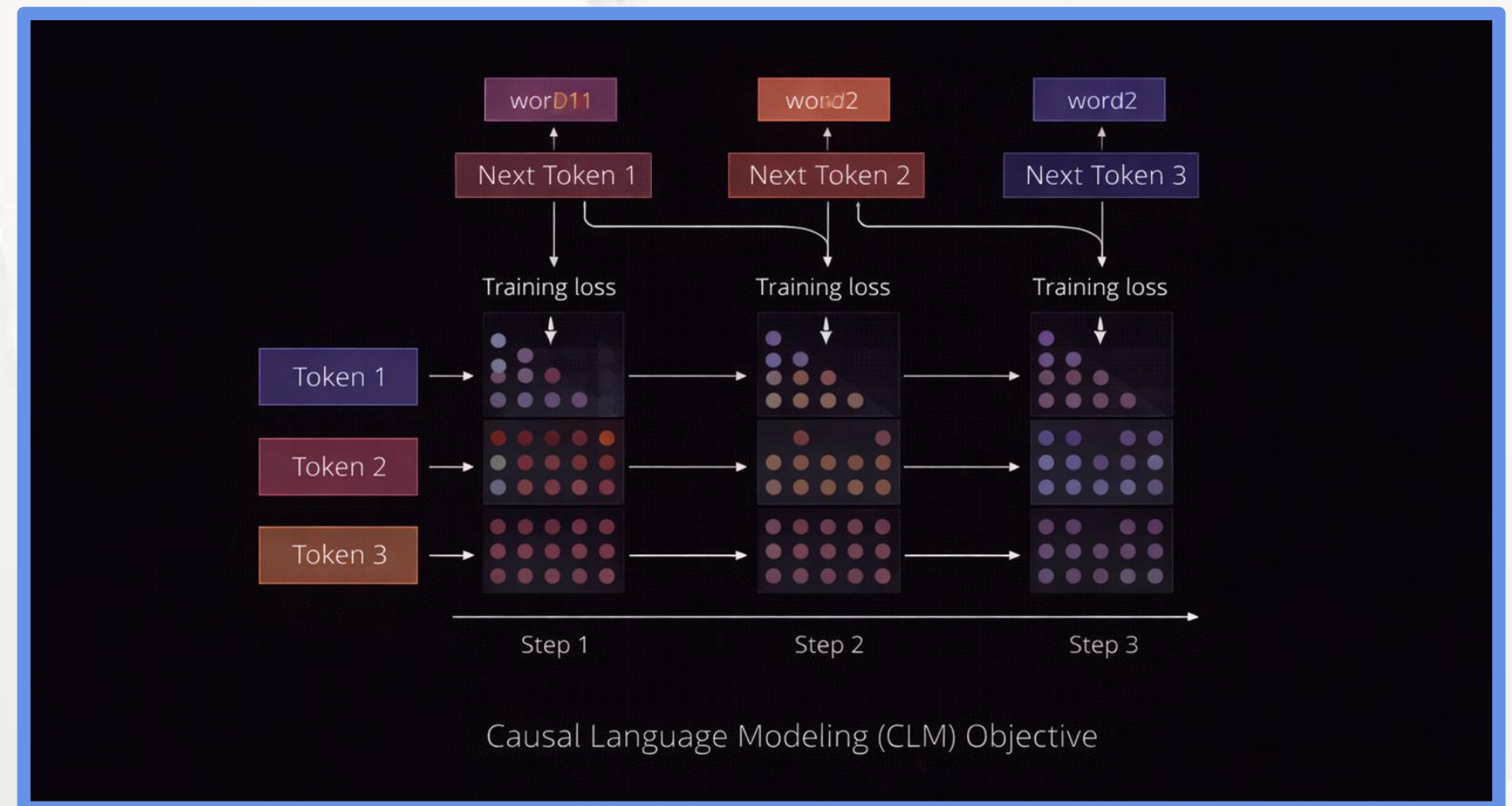
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Chapter 3.5 Decoder-Only Models

GPT Training Objective

- Causal Language Modeling (CLM).
- Predict token $t+1$ given tokens $\leq t$.
- Trained on massive unlabeled text.
- Simple objective, emergent intelligence.
- Model learns from examples in prompt.
- No weight updates required.
- Task defined by context.
- Emergent property of scale.

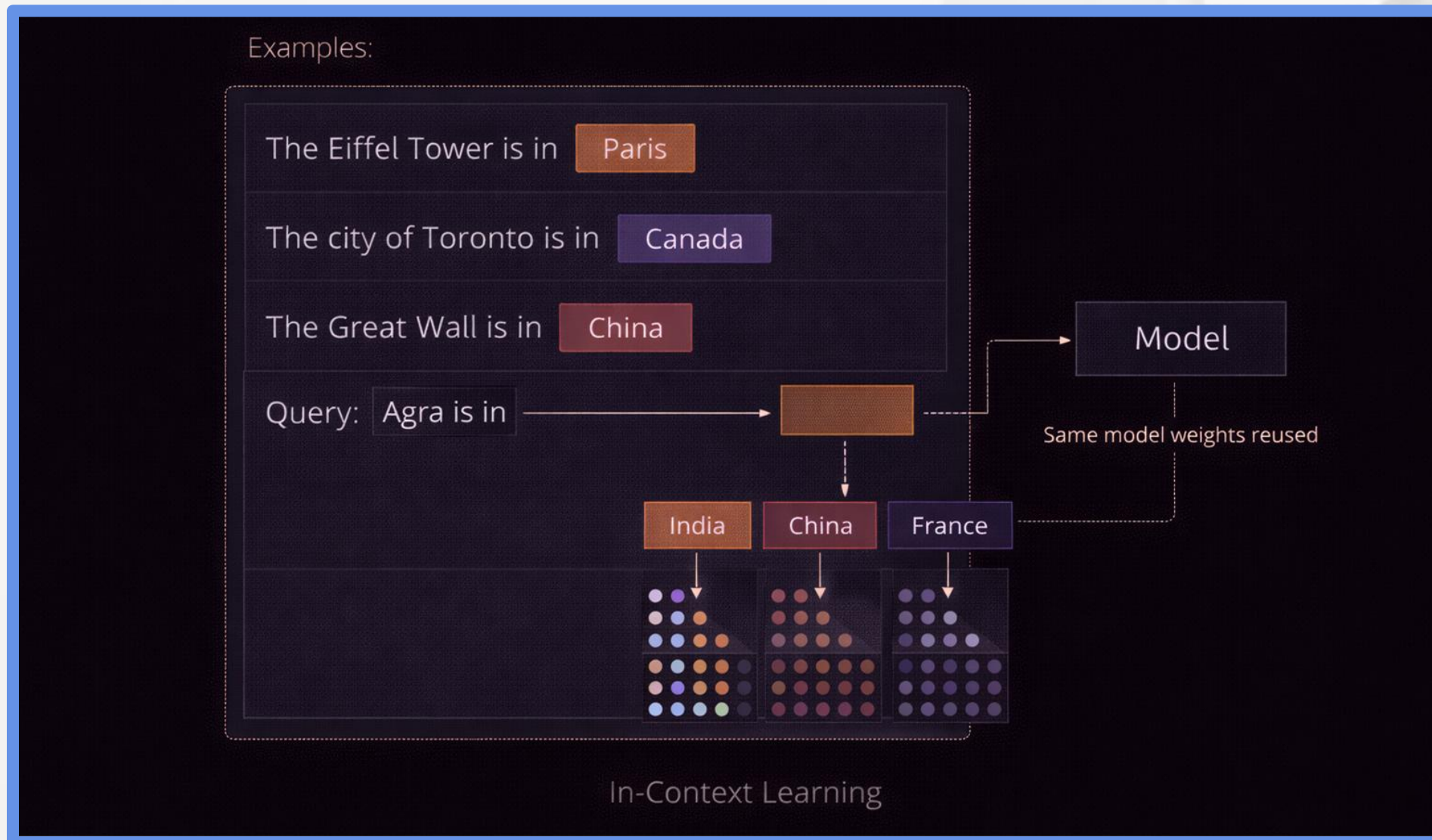


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Chapter 3.5 Decoder-Only Models

Typical Use Cases of Decoder-Only Models



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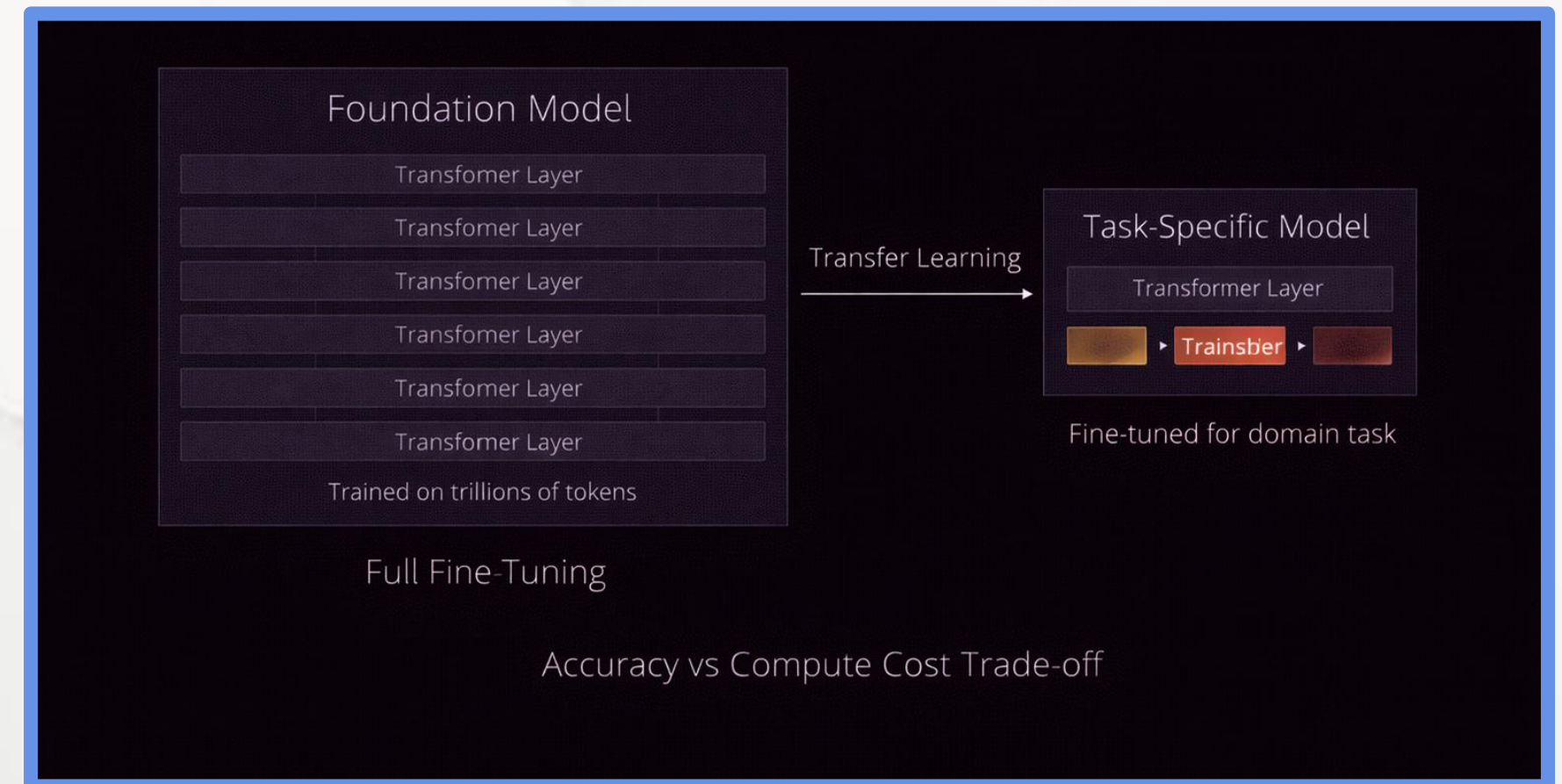
3.7 Summary and The Future of LLMs



Chapter 3.6 Fine-Tuning Strategies & PEFT (LoRA)

What Is Transfer Learning?

- Reuses a large pre-trained foundation model.
- Avoids training from scratch.
- Requires much less data and compute.
- Enables specialization (law, medicine, security).



Chapter 3.6 Fine-Tuning Strategies & PEFT (LoRA)

Full Fine-Tuning vs Parameter-Efficient Fine-Tuning

Two Main Adaptation Paradigms

Full Fine-Tuning

- Updates all model weights
- High accuracy, but high cost

PEFT (Parameter-Efficient Fine-Tuning)

- Updates < **0.1%** of parameters
- Low cost, efficient training
- Faster deployment



Key Trade-Off

Accuracy vs. Training & Deployment Cost

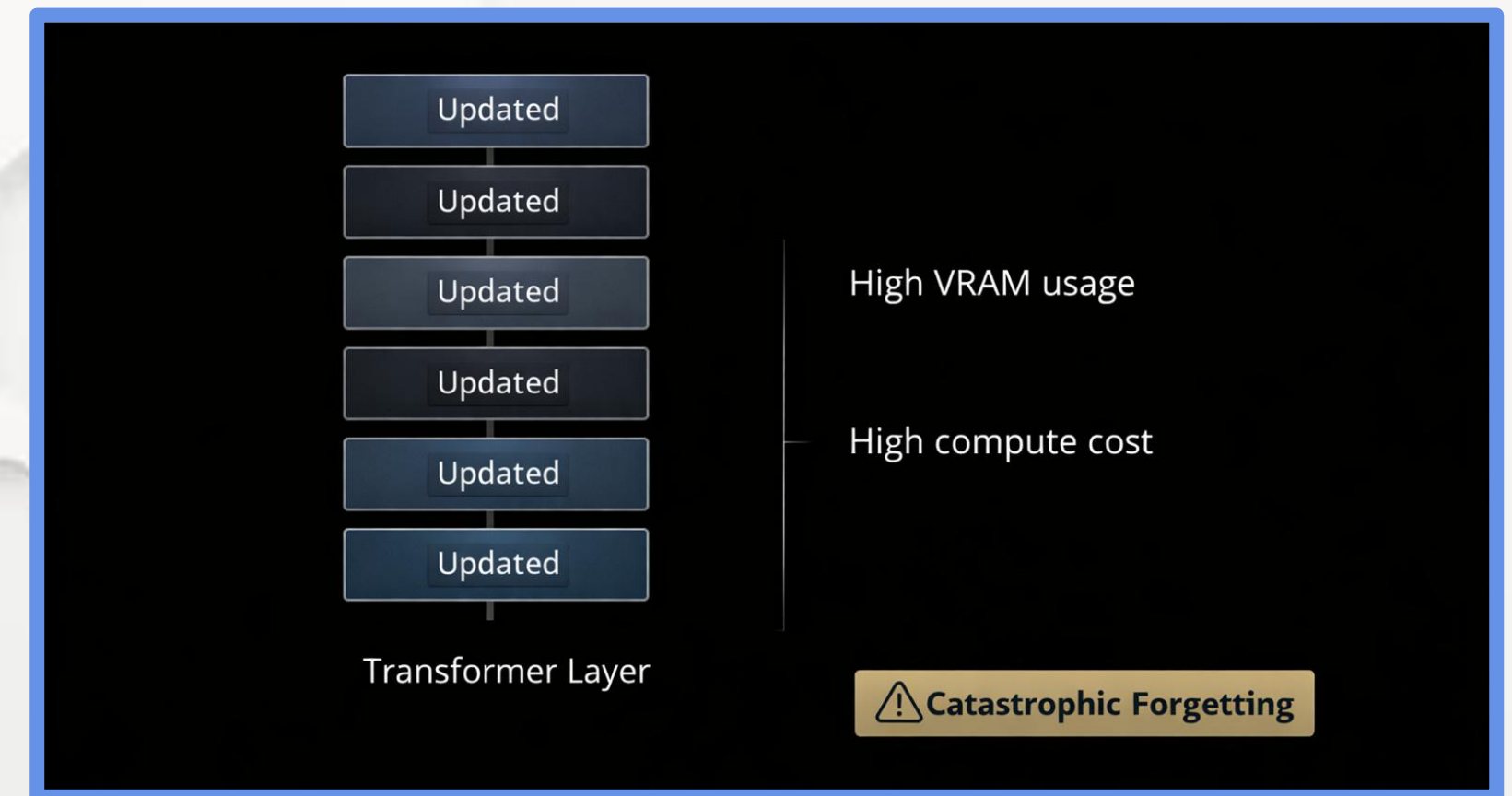


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Chapter 3.6 Fine-Tuning Strategies & PEFT (LoRA)

Full Fine-Tuning: Characteristics

- Updates every parameter in the model.
- Highest adaptation flexibility.
- Very high GPU and VRAM usage.
- Risk of catastrophic forgetting.

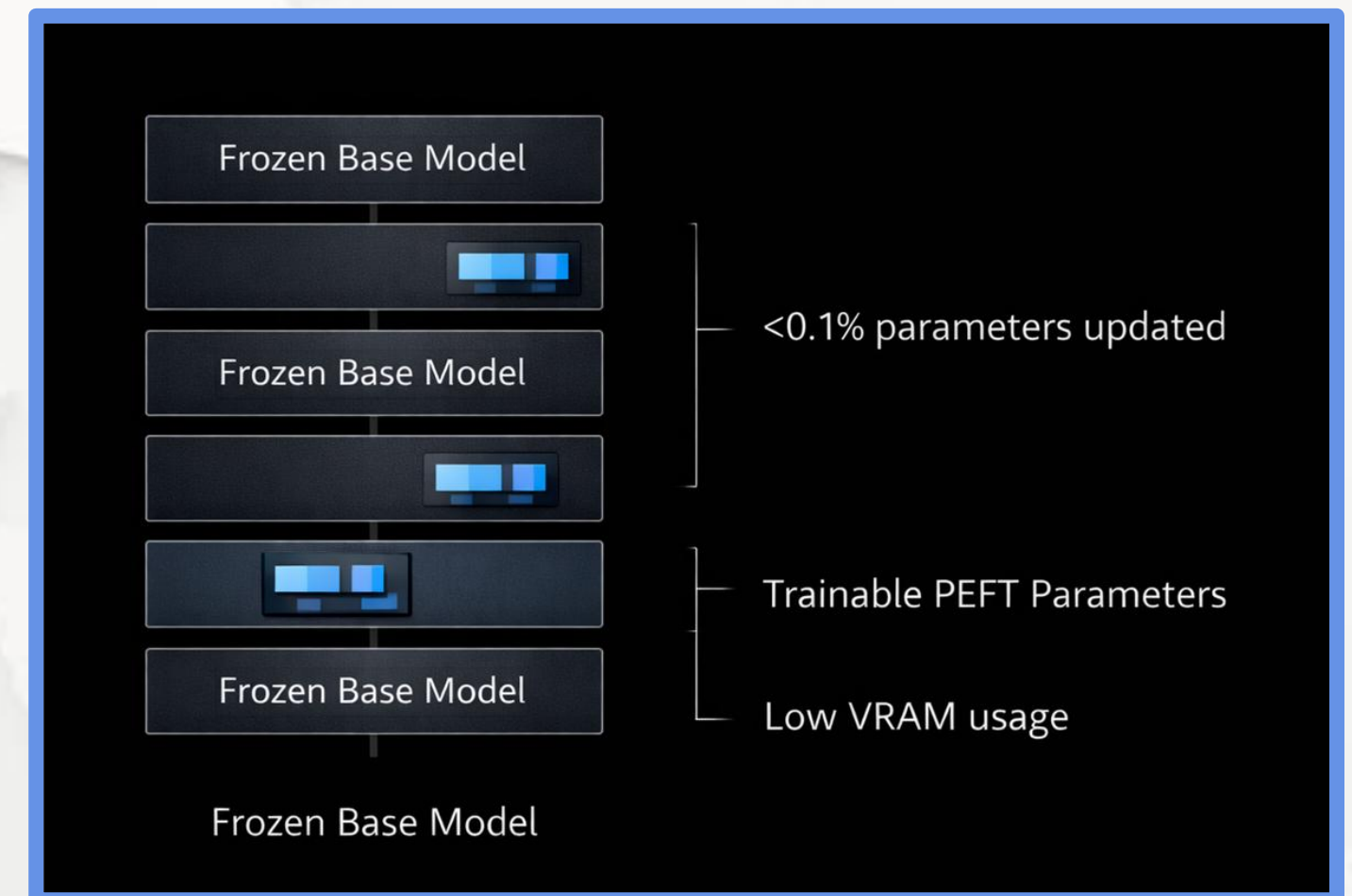


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Chapter 3.6 Fine-Tuning Strategies & PEFT (LoRA)

PEFT: Parameter-Efficient Fine-Tuning

- Base model weights are frozen.
- Only small parameter blocks trained.
- Preserves original model knowledge.
- Much lower memory requirements.



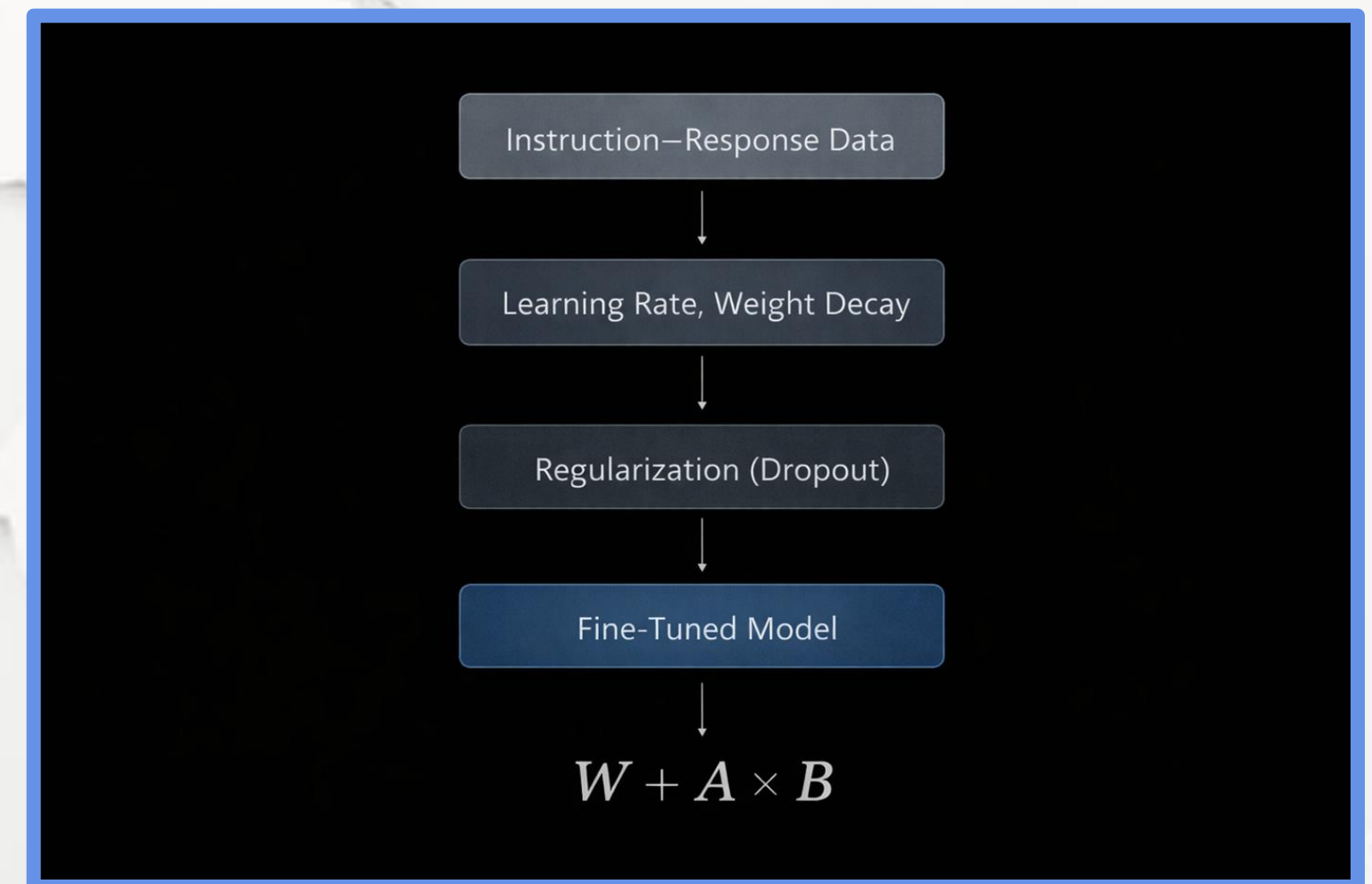
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Chapter 3.6 Fine-Tuning Strategies & PEFT (LoRA)

Choosing the Right Architecture

- Injects low-rank matrices into attention layers.
- Only LoRA parameters are trained.
- Base weights remain unchanged.
- No inference latency penalty.



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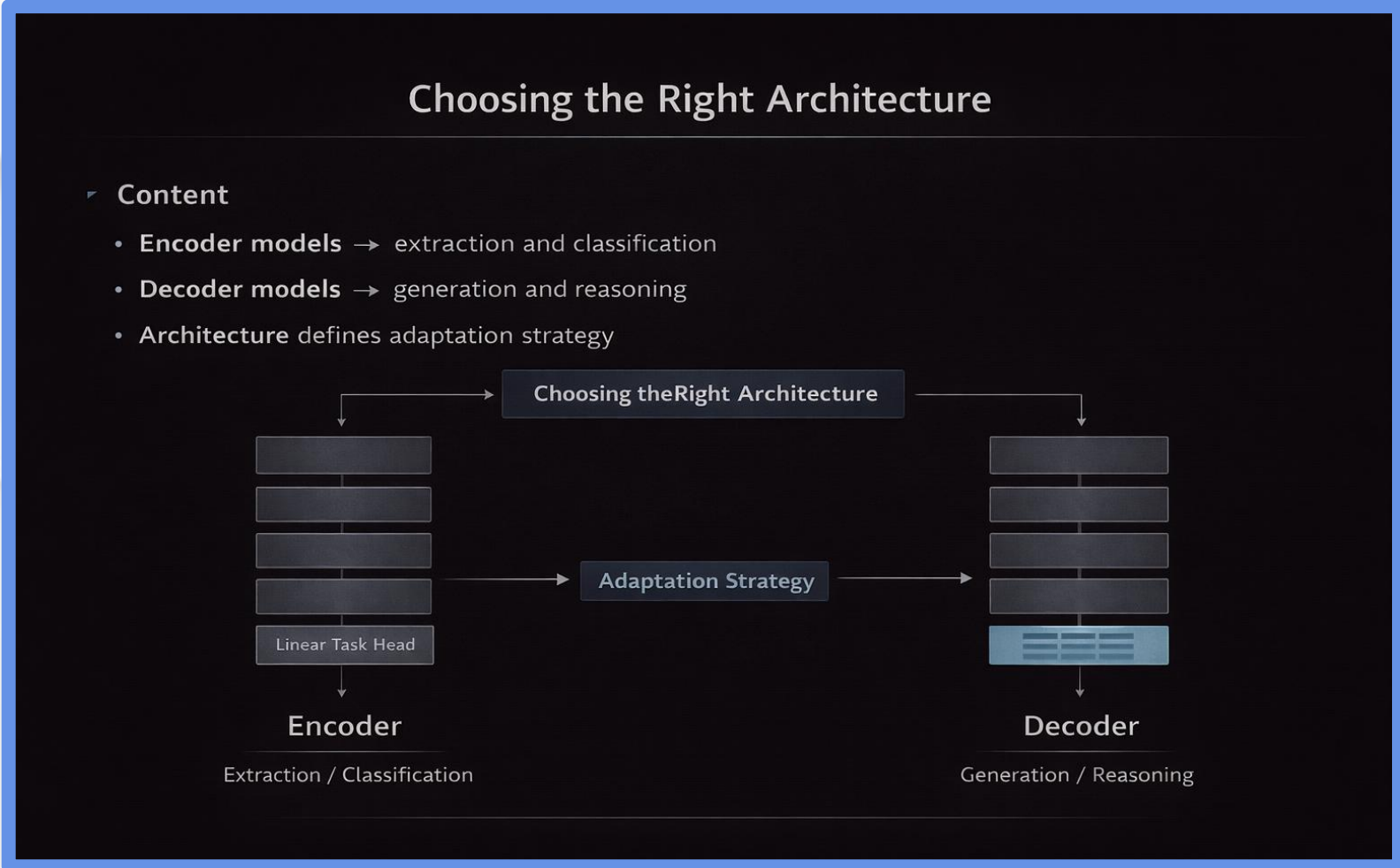


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Chapter 3.7 Summary and The Future of LLMs

Choosing the Right Architecture

- RNNs introduced sequence modeling.
- Transformers removed recurrence limits.
- Attention enabled global context.
- Scale unlocked emergent intelligence.
- Attention is the core computation.
- Encoders specialize in understanding.
- Decoders specialize in generation.
- Architecture defines capabilities.



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Chapter 3.7 Summary and The Future of LLMs

Choosing the Right Architecture

- Vaswani et al. (2017) – Attention Is All You Need.
- Devlin et al. (2018) – BERT: Pre-training of Deep Bidirectional Transformers.
- Brown et al. (2020) – Language Models are Few-Shot Learners.
- Hu et al. (2021) – LoRA: Low-Rank Adaptation of Large Language Models.
- Jay Alammar – The Illustrated Transformer & GPT Series.
- Hugging Face – Transformers & PEFT Documentation.
- Stanford CS224N – Transformer & Attention Lectures.





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